

Fairness in Medical Imaging: Assessment, Mitigation, and Pitfalls

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A. Introduction

Fairness in machine learning has gained increasing importance, particularly in the medical domain where biased decisions may reinforce health disparities. Fairness was investigated in medical imaging using the NIH Chest X-ray dataset, focusing on Pneumothorax detection. The sensitive attribute under consideration is sex (male/female).

This study covers two main aspects: (1) fairness assessment and mitigation using the `Fairlearn` library, and (2) pitfalls arising from label distortion. Through this, the aim is to highlight both the benefits and the limitations of fairness interventions in real-world scenarios.

B. Dataset & Setup

The dataset used is the NIH Chest X-ray dataset, containing 18,326 images. In the project, the target label to assess for is Pneumothorax. To ensure comparability, all images were preprocessed to 224×224 pixels as is custom for the ResNet structure, and a balanced male/female split was constructed; both in testing and validation.

A ResNet architecture serves as the classification model. The ResNet is loaded in from a checkpoint after having already be trained.

C. Fairness Assessment (Part 1)

The baseline model was first evaluated without mitigation. Group fairness metrics are computed: accuracy, true positive rate (TPR), false positive rate (FPR), and selection rate across male and female groups. These allow for the assessment of the fairness metrics of *Independence*, *Seperation* and *Sufficiency*.

Plots of the ROC curve for Part 1 were made to visualize group disparities, but were omitted from the report in favor of other plots. Refer to the Python notebook in annexes.

Fairness is interpreted through multiple lenses:

- **Statistical Parity:** are predictions independent of sex? I.e., $P(\hat{Y} = 1 \mid \text{Sex} = M) = P(\hat{Y} = 1 \mid \text{Sex} = F)$

- **Equal Opportunity:** are TPRs equal across groups? I.e., $P(\hat{Y} = 1 \mid Y = 1, \text{Sex} = M) = P(\hat{Y} = 1 \mid Y = 1, \text{Sex} = F)$

- **Separation:** do both TPR and FPR align across groups? I.e., $P(\hat{Y} = 1 \mid Y = 0, \text{Sex} = M) = P(\hat{Y} = 0 \mid Y = 1, \text{Sex} = F)$

In this report we focus on reducing difference in separation score and thus only report this.

D. Bias Mitigation

Threshold Optimization was applied, a post-processing method that adjusts decision thresholds per group. Table 1 reports fairness metrics before and after mitigation. It is possible to optimize for different metrics, and in this report, we optimize for balanced *separation*.

For each class, the new threshold function becomes $p_{\text{ignore}} \cdot c + (1 - p_{\text{ignore}}) \cdot (p_0 \cdot \text{operation}_0(\text{score}) + p_1 \cdot \text{operation}_1(\text{score}))$, where p_{ignore} , p_0 , and p_1 are learned by the Threshold Optimizer.

	Accuracy		Seperation	
	Male	Female	Male	Female
Before	0.85	0.77	0.56	0.62
After	0.64	0.67	0.74	0.71

Table 1. Fairness metrics before vs. after threshold optimization.

E. Pitfall: Label Distortion (Part 2)

It is uncertain that the test labels will always correspond to the ground truth. For example, an image from the data set may be labelled as "healthy" when in reality, the image is of a diseased patient (perhaps the diagnosis was missed). To investigate the effect of distorted labels on our bias mitigation strategy, the testing was repeated from *C. Fairness Assessment Methodology*, simulating label distortion by flipping 30% of healthy male labels to "Pneumothorax" in our test and validation sets. Figures 1-3 and Table 2 illustrate our findings before mitigation and after mitigation (both with respect to the distorted labels and original, undistorted labels).

	Accuracy		Seperation	
	Male	Female	Male	Female
Dist. Labels	0.60	0.76	0.30	0.27
Or. Labels	0.75	0.76	0.72	0.27

Table 2. Fairness metrics after threshold optimization evaluated on distorted labels and orginal labels

F. Discussion

Looking at the results in Table 1 we see that the Threshold Optimization successfully decreased the difference in separation score from a difference of 0.05 to a difference of 0.02, while at the same time bringing accuracy closer together. This, however, came with a cost of significantly lower accuracy for *both* sex-group. Thus, by seemingly making the model fairer, we lowered performance for both groups, thus begging the question of whether this approach is favourable at all.

Comparing these results to the procedure where a percentage of the labels of the test and validation sets were distorted, a stark difference can be noted in both metrics (See Table 2) and in the ROC curves (see figure 1 and figure 2). First, we note that the ROC curve for the male group is far more close to a random classifier, compared to the female group. After mitigation, the equalized odds are very promising on the distorted data, but for *undistorted* data, it introduced a worse fairness measure than originally present (See table 2). This introduces a challenge with fairness mitigation if the data in question is not thoroughly sanitized.

Through experimentation, it became clear that the outcome of threshold optimization was highly sensitive to which samples had their labels flipped, suggesting that multiple runs are necessary to properly account for this variation.

G. Conclusion

In conclusion, this project shows that fairness interventions like threshold optimization can reduce disparities between groups but often lower overall accuracy. The label distortion experiment further revealed how strongly fairness outcomes depend on data quality, emphasizing the need to balance fairness, performance, and reliable datasets when applying AI in critical domains such as healthcare.

Annexes

For additional materials and code, see the project repository.

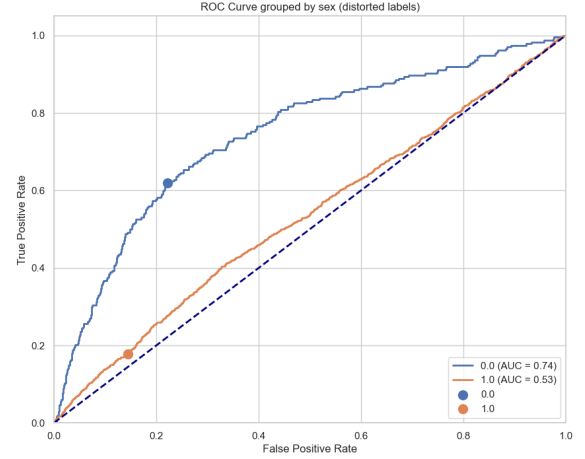


Figure 1. ROC curves for male vs. female groups before mitigation (distorted labels).

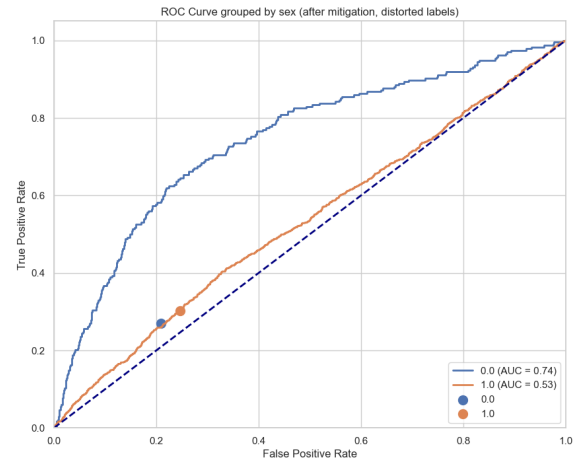


Figure 2. ROC curves for male vs. female groups after mitigation (distorted labels).

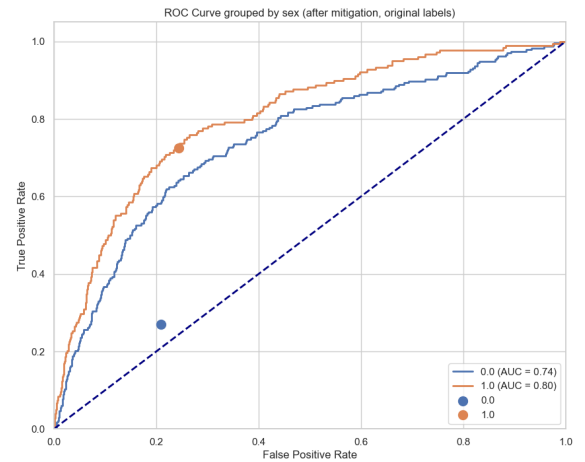


Figure 3. ROC curves for male vs. female groups after mitigation (original labels).